

Security Risk Assessment Report

Network Hardening — Social Media Organization Data Breach

Part 1: Select up to three hardening tools and methods to implement

An inspection of the organization's network following the recent data breach revealed four vulnerabilities: employees share passwords, the admin password for the database is set to the default, the firewalls do not have rules to filter traffic coming in and out of the network, and multifactor authentication (MFA) is not used. To address these consistently and prevent another breach, the following three network hardening tools and methods are recommended: multifactor authentication (MFA), password policies, and port filtering. The table below maps each vulnerability to the hardening method that addresses it.

Vulnerability	Hardening tool / method
Employees share passwords	Network access privileges; password policies
Admin database password is set to the default	Multifactor authentication (MFA); password policies
Firewalls have no rules to filter traffic in and out of the network	Port filtering; firewall maintenance
Multifactor authentication (MFA) is not used	Multifactor authentication (MFA)

These three were selected because they cover all four vulnerabilities, and two of them address more than one weakness at once. MFA directly resolves the missing-MFA vulnerability and strengthens the default-password vulnerability. Password policies support both the shared-password and default-password issues. Port filtering closes the gap created by firewalls that lack traffic-filtering rules. Network access privileges are also noted alongside password policies as a supporting measure for the shared-password vulnerability.

Part 2: Recommendations

Two recommendations are explained in detail below: multifactor authentication and port filtering. Together they address the authentication weaknesses and the network traffic weakness that made the breach possible.

- **Multifactor authentication (MFA).** MFA requires a user to verify their identity in two or more ways before accessing a system or network, using factors such as a password, a PIN, a one-time password (OTP) sent to a cell phone, a badge, or a fingerprint. It is effective here because it helps protect against brute force attacks and similar events: even if an attacker obtains or guesses a password, such as the default admin password on the database, they would still need a second factor to gain access. This addresses two of the organization's vulnerabilities at once, the unused MFA and the default admin password. In terms of frequency, MFA is mostly a

technique that is set up once and then maintained, rather than something performed on a recurring schedule, which makes it a high-value control for the effort involved.

- **Port filtering.** Port filtering is a firewall function that blocks or allows certain port numbers to limit unwanted communication. It is effective because it controls network traffic and can prevent potential attackers from entering the private network, directly addressing the vulnerability that the firewalls currently have no rules to filter traffic coming in and out of the network. By allowing only the ports the organization actually needs and blocking the rest, the attack surface is reduced. This is paired well with regular firewall maintenance, which involves checking and updating security configurations regularly to stay ahead of potential threats and can be performed on an ongoing basis, so that filtering rules remain current as the network changes.

Implemented together, these methods harden the network against the specific weaknesses identified in the assessment: password policies and MFA strengthen authentication so credentials alone are not enough to gain access, network access privileges limit what shared or unauthorized accounts can reach, and port filtering with firewall maintenance controls the traffic entering and leaving the network. The more of these safeguards that are in place, the less likely a malicious actor will be able to access sensitive information.